Charles Goodsir

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Auckland, NZ | Open to roles in London or Melbourne

PROFESSIONAL SUMMARY

Application engineer with 5+ years delivering enterprise CRM/ERP systems for NZ and AU councils, now moving into application security and DevOps. I build and then attack my own systems: a gated CI/CD pipeline (Semgrep, gitleaks, Trivy, OWASP ZAP) around a vulnerable-by-design app, and a Terraform-managed Azure landing zone that deploys only an approved, scanned plan. Security+ certified, PortSwigger SQLi and XSS paths complete. NZ and UK citizen with full Australian and UK work rights, no visa needed.

WORK EXPERIENCE

Datacom Solutions

Sep 2021 - Present

Application Engineer

Auckland, NZ

- Review every migration before it merges to confirm it does what it should, and caught several that would have overwritten customer data and configuration and forced councils back to earlier versions.
- Turned a change in resource management law into 5 system changes and delivered them on my own for the fixed 30 October 2026 go-live: a new section 104AA notice pathway, updated planning application processes, a new Permitted Activities report and revised Permitted Activities checklists, keeping resource consent processing compliant under the transitional rules for the councils that use it.
- Diagnosed a creditor invoice workflow that had halted invoice processing for a sprint, traced it to incorrect mappings in a workflow expression, and fixed it in live.
- Found a misconfigured condition that left planning-application milestones neither completing nor notifying, so applications went overdue with no escalation, and corrected it in live.
- Built incident logging so the correct contacts are recorded and officers are notified, including raising incidents directly from Compliance Monitoring Surveys with the survey's locations carried into the new record.
- Built dashboards for council teams to see their own data on planning applications, RFIs and expiring applications.
- Build local development environments with test data and configuration, then export the configuration as JSON files used to create or update council environments. Built hotfix environments for urgent council issues.
- Authored YAML-driven test-data configuration in Azure DevOps, layered on top of an existing C# codebase, to provision cross-referenced data for local and cloud environments at build time.
- Helped cut configuration defect rates by ~30% for NZ & AU council clients by standardising testing.

Datacom Solutions (MBIE)

Sep 2020 - Sep 2021

Service Desk Analyst

Wellington, NZ

- Resolved 200+ IT incidents a month for the Ministry of Business, Innovation and Employment, with first-call resolution above 80%.
- Cut repeat support volume by 25% by spotting recurring issues and writing knowledge-base articles the wider team adopted.
- Triaged critical infrastructure incidents (network access, identity management, remote desktop) and escalated them to specialist teams within SLA.

SKILLS

Security: Semgrep (SAST), gitleaks (secrets), Trivy (container/IaC scanning), OWASP ZAP (DAST), Dependabot (SCA), Burp Suite, OWASP Top 10, secure CI/CD pipeline design, STRIDE threat modelling, NIST CSF 2.0, CSP & security headers, Entra ID & OAuth 2.0, Microsoft Sentinel (KQL), Defender for Cloud, Azure Policy, threat-oriented UAT

Cloud & Infra: Azure, Azure DevOps (Pipelines, self-hosted agents), Terraform, Docker, Linux, nginx, AWS (Lambda, API Gateway, SAM, DynamoDB, EventBridge, CloudWatch, Lightsail), Cloudflare Workers + KV, .NET tooling, Supabase (Postgres + PostGIS), Firestore, CI/CD (Azure Pipelines, GitHub Actions, Expo EAS), Git

Languages: TypeScript, JavaScript, Python, SQL, Bash, YAML/JSON (config authoring), HTML/CSS. C# and ASP.NET Core (an API written as a guided code-along project); otherwise works within existing C# codebases via configuration.

Frameworks: React Native, React, Node.js, ASP.NET Core, FastAPI **Tools:** Confluence, Jira, Miro

PROJECTS

Secure Expense Claims Complete

Wrote a STRIDE threat model of 24 threats before any code, then built an expense claims system (ASP.NET Core API, React, Postgres) with ownership checks in every query, separation of duties, and 65 API tests against real Postgres. Attacked my own access control with 28 attacks using real Entra ID tokens for each role (IDOR, self-approval, tampering, forged tokens). None got through; the run found an Entra setting on the wrong app, which I fixed in two layers. Deployed to Azure with no secrets (OIDC pipeline, managed identity, read-only plan identity, approval before every apply, weekly drift check) and gated every PR on 8 required checks, breaking each control on purpose to prove its test fails.

ASP.NET Core, React, TypeScript, Postgres, Terraform, Azure (Entra ID, Policy, Defender, Sentinel), GitHub Actions, OIDC. Built as a guided code-along.

appsec-homelab Complete

Built a vulnerable-by-design ASP.NET Core 8 / React app with four seeded vulnerabilities (SQL injection x2, reflected XSS, plaintext passwords), then exploited, fixed, re-tested and documented each one. Parameterized queries and salted PBKDF2 hashing replaced the flaws, verified with manual exploitation and regression tests. Built a CI/CD security pipeline with gitleaks, Semgrep, Dependabot, Trivy, OWASP ZAP and a reviewer-approval gate before production, moved from GitHub Actions to Azure Pipelines with a self-hosted agent. Hardened the nginx config over three ZAP scan rounds (CSP, X-Frame-Options, Permissions-Policy, cross-origin isolation headers), cutting warnings from 8 to 3, with the remaining trade-off documented.

ASP.NET Core 8, React, nginx, Docker, GitHub Actions, Azure Pipelines, Semgrep, gitleaks, Trivy, OWASP ZAP

Secure Azure Landing Zone Complete

Built a Terraform-managed Azure landing zone (VNet, deny-by-default NSG, storage, Key Vault with RBAC and purge protection) through a 5-stage Azure DevOps pipeline: validate, security scan, plan, manual approval, then apply only the approved saved plan. Authenticated with workload identity federation (OIDC), so there is no stored secret, and added weekly drift detection that fails when Azure no longer matches the code. Triaged 6 scanner findings on my own Terraform: fixed 4, including a CRITICAL Key Vault with no network ACL, and accepted 2 with a written reason and an expiry date.

Terraform, Azure, Azure DevOps, Trivy, OIDC, YAML

PortSwigger Web Security Academy (Burp Suite Community) In progress

Completed the SQL injection (including blind, with Python solver scripts) and cross-site scripting (including AngularJS sandbox escapes and CSP bypasses) learning paths.

Burp Suite, Python

Detour: Cross-Platform Mobile App (Solo Developer) Live on App Store

Built a full-stack mobile app with auth, geolocation, real-time data, and social features, and implemented Supabase Auth with row-level security policies restricting each user to their own data. Got geolocation queries under 200ms by optimising PostGIS spatial indexes on a self-managed Postgres instance.

React Native, TypeScript, Supabase, Firestore, PostGIS

EDUCATION

Level 6 Diploma in Applied Software Development Dev Academy Aotearoa 2023 - 2024

BA in History, International Relations & Political Science Victoria University of Wellington 2017 - 2020

CERTIFICATIONS

CompTIA Security+ (SY0-701) CompTIA Jul 2026